USED-CAR RESALE PRICE PROJECT

Two-Stage Data Dictionary | Phase 1: Data Understanding

> **Dataset Source:** Kaggle (sujay1844/used-car-prices), File: train.csv
> **Dimensions:** 5,847 rows × 14 columns | **Target:** Price

1. Raw (Original) Data Dictionary

Created before cleaning. It records source column names, meanings, raw type/role, and known quirks.

Original Column	Meaning	Raw Type / Role	Known Notes / Quirks
Unnamed: 0	Row identifier included in source file	Integer / identifier	Row index or ID; not a vehicle characteristic.
Name	Car make and model as listed	Categorical text	Contains listed brand and model.
Location	City in which used car is listed	Categorical text	City name.
Year	Year car was manufactured	Integer / numerical	Calendar year; renamed during preparation.
Kilometers Driven	Distance the car has travelled	Numerical	Kilometres; one extreme value of 6,500,000 flagged in notes.
Fuel Type	Fuel category of the vehicle	Categorical text	Examples: Diesel, Petrol, Electric.
Transmission	Transmission system of vehicle	Categorical text	Examples: Manual, Automatic.
Owner_Type	Ownership category in listing	Categorical text	First, Second, Third, Fourth & Above.
Mileage	Reported fuel-efficiency figure	Unit-bearing text	Extract numeric portion; verify raw suffixes before assigning universal unit.
Engine	Engine displacement/capacity	Unit-bearing text	Numeric portion extracted; unit is CC.
Power	Reported engine power	Unit-bearing text	Numeric portion extracted; unit is bhp.
Seats	Number of seats in vehicle	Numerical / integer-valued	Missing values possible; nullable integer used after prep.
New Price	Reported new/showroom price	Text / object	5,032 missing out of 5,847; renamed Showroom_Price then dropped.
Price	Listed resale price	Numerical / target	Notebook plots label in lakh; exact source currency not confirmed.

Raw-data summary: *Dataset contains 1,804 car models, 11 locations, 22 years of data, 3 fuel types, 2 transmission types, and 4 owner categories. No duplicate rows reported. New_Price contains 5,032 missing values.*

2. Clean (Final) Data Dictionary

Describes the prepared schema and actions documented in the supplied materials. Uploaded Jupyter notebook contains data loading and initial understanding, but not cleaning code.

Final Column	Meaning	Clean Type / Role	Preparation / Final Notes
Name	Car make and model	String / categorical	Retained; converted to string type.
Location	City where car is listed	String / categorical	Retained; converted to string type.
Manufacturing_Year	Year of manufacture	Integer / numerical	Renamed from Year.
Kilometers Driven	Distance travelled	Numerical	Retained; 6,500,000 treated as data-entry error and replaced with Name median.
Fuel_Type	Fuel category	String / categorical	Retained; converted to string type.
Transmission	Transmission system	String / categorical	Retained; converted to string type.
Owner Type	Ownership category	String / categorical	Retained; converted to string type.
Mileage	Fuel-efficiency figure	Float / numerical	Numeric portion extracted; 40 zero values changed to missing. Imputed by Name, then Fuel_Type median. Electric vehicles may remain missing.
Engine	Engine capacity	Float / numerical	Numeric portion extracted (CC). Imputed by Name, then Fuel_Type median.
Power	Engine power	Float / numerical	Numeric portion extracted (bhp). Imputed by Name, then Fuel_Type median.
Seats	Number of seats	Nullable integer	Missing values filled by Name mode, then overall mode.
Price	Resale / listing price	Numerical / target	Retained as dependent variable. 679 statistical outliers flagged via 1.5 × IQR rule; not removed.

3. Column-Level Changes

- **Dropped:** `Unnamed: 0` (row identifier).
- **Renamed:** `Year` → `Manufacturing_Year`; `New_Price` → `Showroom_Price` before being dropped.
- **Type Conversions:** Name, Location, Fuel_Type, Transmission, and Owner_Type to string; Mileage, Engine, Power numeric portions to float; Seats to nullable integer. Final retained: 11 features + Price target (12 columns total).

4. Data-Quality Handling and Caveats

- Showroom_Price has 5,032 missing values and is dropped.
- 40 zero values in Mileage converted to missing prior to median imputation by Name and Fuel_Type.
- Non-electric Engine and Power missing values filled using medians grouped by Name and Fuel_Type.
- One extreme Kilometers_Driven value of 6,500,000 replaced with car-Name median.
- 679 Price observations flagged as outliers using 1.5 × IQR rule; retained in dataset.

5. Target, Features and Source Uncertainties

Target: Price. **Potential Features:** Name, Location, Manufacturing_Year, Kilometers_Driven, Fuel_Type, Transmission, Owner_Type, Mileage, Engine, Power, Seats. Notebook plots label Price in lakh, but exact currency requires verification against raw metadata.